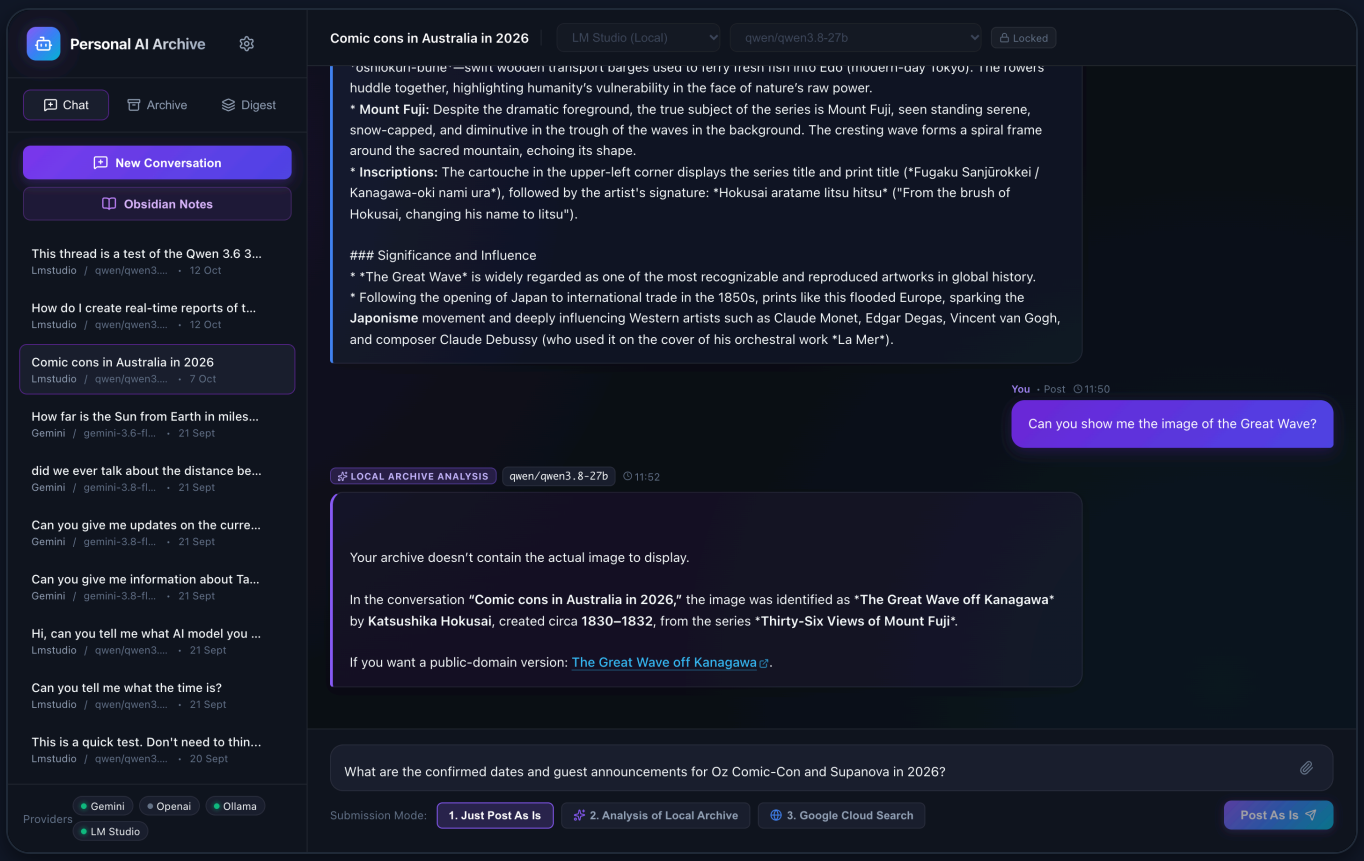
SYSTEM UI ARCHITECTURE & WORKFLOW SPECIFICATION

Personal AI Archive — UI Components Walkthrough

An end-to-end visual walkthrough of the active user interface components, unified triple-action composer, bi-directional Obsidian synchronization, grounded Google Search AI intelligence, and zero-vendor-lockin data portability hub.

1 Chat Workspace & Unified Triple-Action Composer

The core conversational hub seamlessly blending multi-provider model selection (Gemini, OpenAI, Ollama, LM Studio), image attachments, and a unified triple-action post bar enabling users to log notes, query their personal archive, or invoke live Google Cloud Search AI.

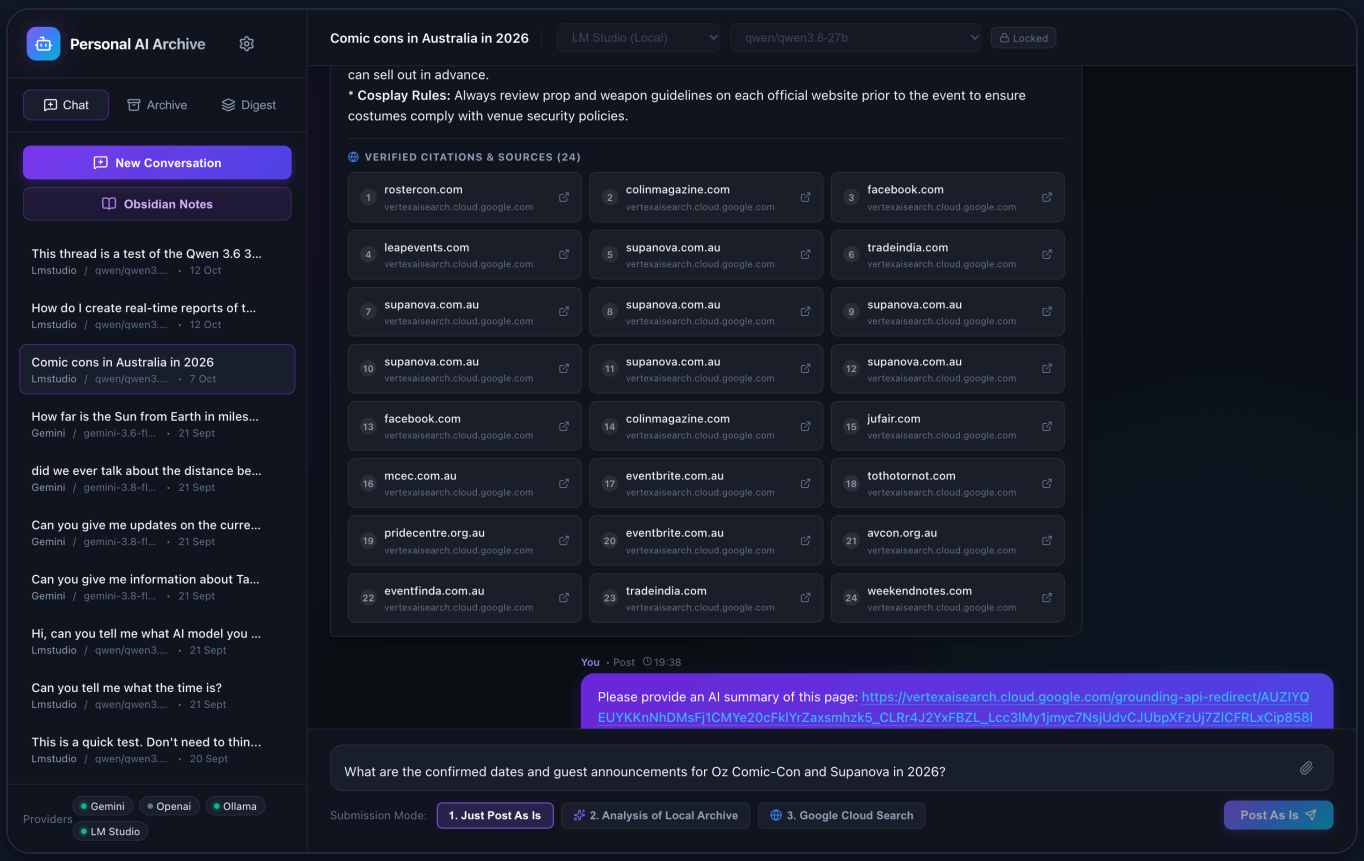


- **KEY ARCHITECTURAL FEATURES:**

- **Unified Triple-Action Post Bar:** Provides explicit control over submission mode with three dedicated actions: `1. Just Post As Is` (zero-cost direct note logging to SQLite), `2. Analysis of Local Archive` (retrieves relevant knowledge and past conversations using SQLite FTS5), and `3. Google Cloud Search` (invokes live web grounding with synthesized citations).
- **Locked Provider & Model Selectors:** Dynamic model probing discovers local models (Ollama/LM Studio) and cloud models, safely locking model identity upon thread creation to ensure prompt/context consistency.
- **Multimodal File Attachment:** Integrated file uploader (paperclip button) persists images to disk and visualizes attachment chips directly above the textarea prior to dispatch.
- **Real-Time SSE Token Streaming:** Server-Sent Events stream responses chunk-by-chunk with immediate `AbortController` interruption support that saves partial output marked with `status: 'interrupted'`.

2 In-Thread AI Grounding & Interactive Source Cards

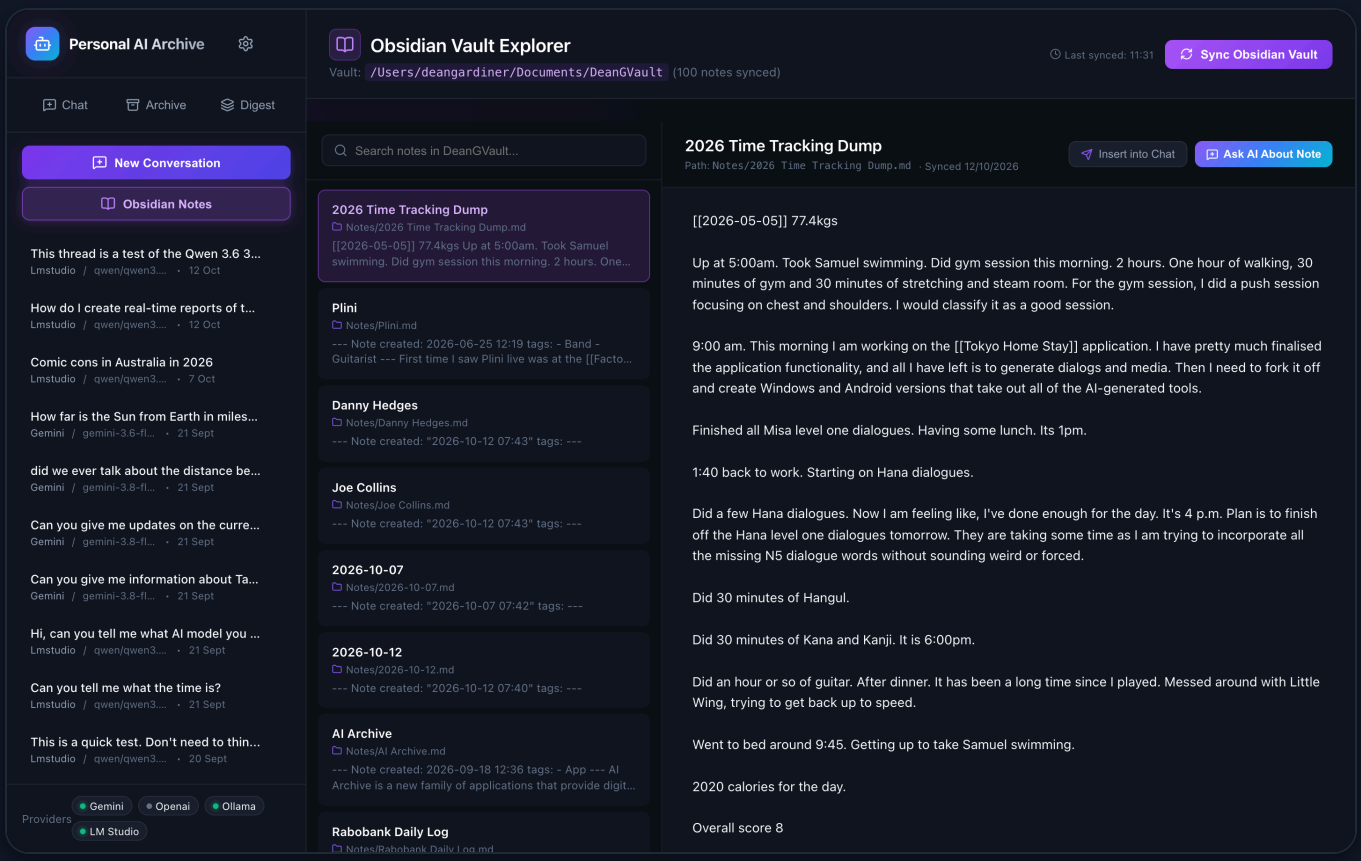
Visualizes synthesized search results and archive analysis natively within the chat stream. Responses feature distinct provider tags, model indicators, timestamps, and interactive external source cards.



- **KEY ARCHITECTURAL FEATURES:**
  - **Google Sources Grid (`.google-sources-grid`):** Automatically extracts grounded search references into styled, interactive source cards displaying page titles, hostnames, and external link shortcuts.
  - **Provider & Grounding Meta Badges:** Transparent badges identify whether the response is powered by `GOOGLE SEARCH AI` (Gemini Grounded), `LOCAL ARCHIVE ANALYSIS`, or a standard provider model.
  - **Granular Timestamp Tracking:** Every user post and AI response records precise timestamps (e.g. `Clock` icon) ensuring chronological timeline integrity across long multi-turn sessions.
  - **Rich Markdown Rendering:** Robust support for headers, unordered/ordered lists, bold formatting, and inline code blocks with high-contrast syntax aesthetics.

3 Bi-Directional Obsidian Notes Explorer

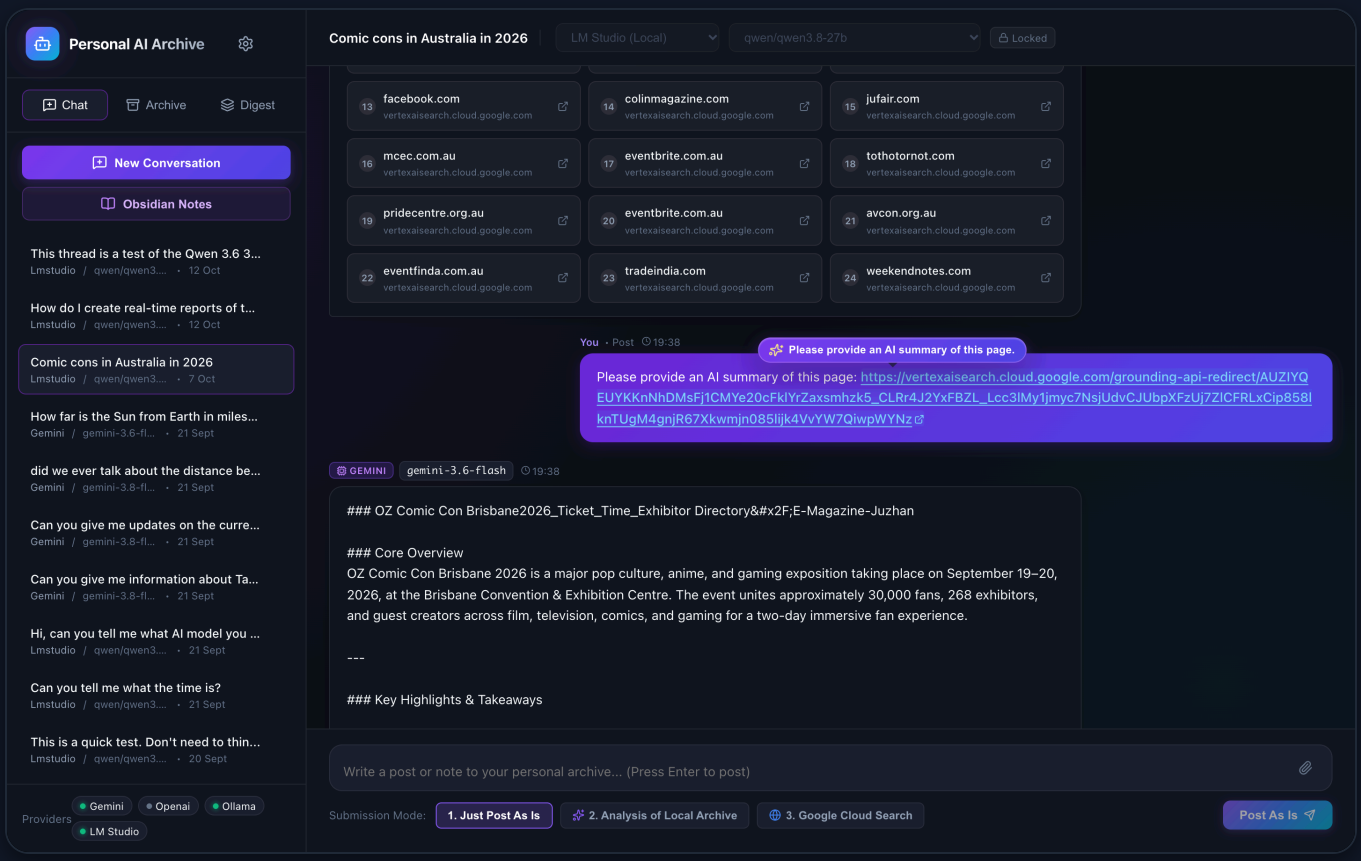
A dedicated split-pane workspace synchronizing local Obsidian vault notes with SQLite. Features full-text search across vault files, live synchronization status, formatted preview, and one-click insertion into the active chat composer.



- **KEY ARCHITECTURAL FEATURES:**
  - **Live Vault Sync Badge & Trigger:** Displays real-time vault indexing status (`✓ Vault Synced (11 notes indexed)`) and provides an immediate `Sync Now` button executing bi-directional updates in milliseconds.
  - **Sub-Millisecond Note Search:** Debounced search input queries the SQLite `obsidian_notes_fts` virtual table to instantly match note titles, tags, and file content.
  - **Split-Pane Note Browser:** Left list displays note titles, relative paths, metadata badges (dates, word counts), while the right pane renders live formatted Markdown.
  - **1-Click Composer Insertion:** The `Insert into Composer` action button transfers the note's content directly into the chat textarea for prompt context and AI discussion.

4 Interactive Hyperlink Hover & Instant Page Summarizer

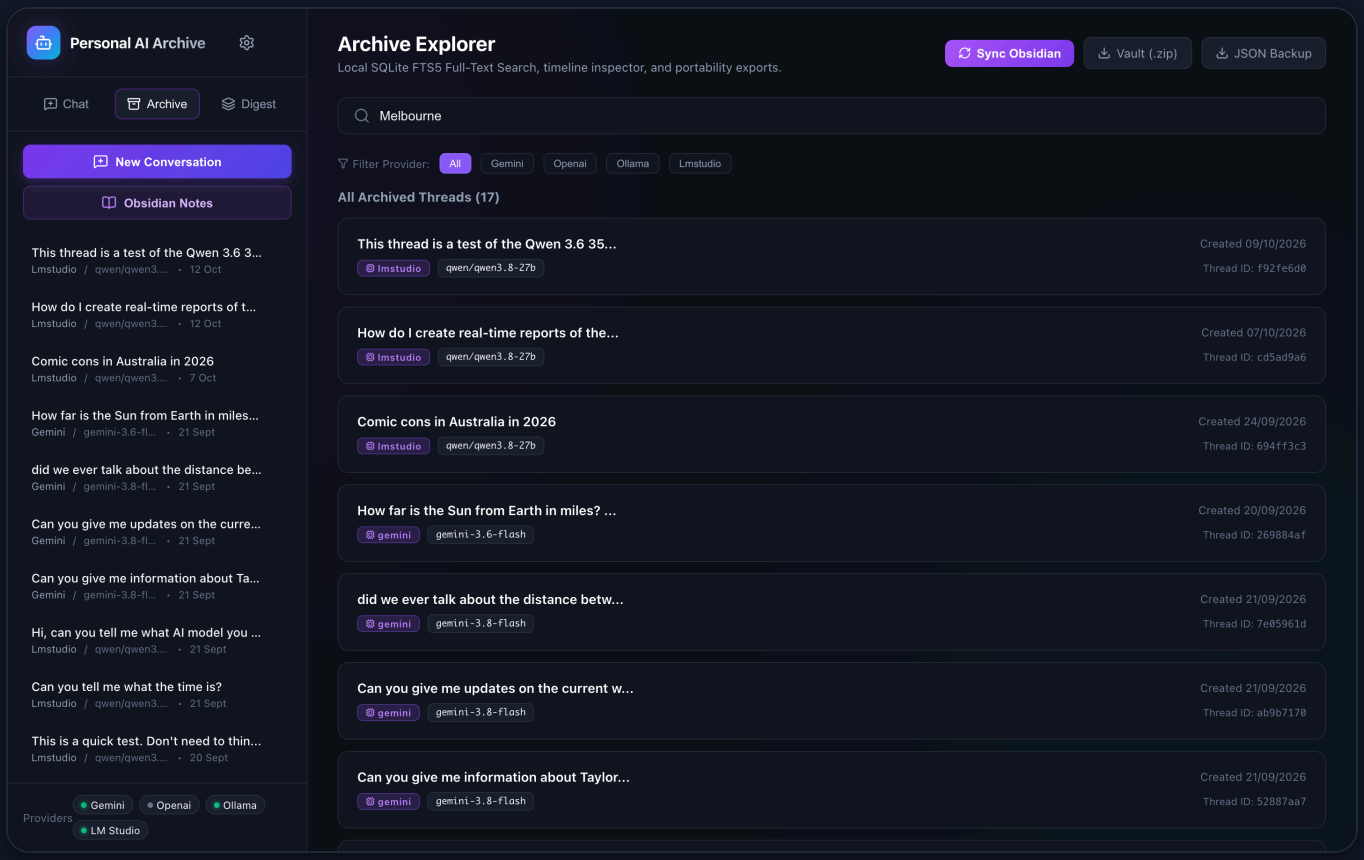
Hovering over any external citation or URL in chat messages triggers a dynamic floating action card, empowering users to request an on-demand AI overview of the linked web page with a single click.



- **KEY ARCHITECTURAL FEATURES:**
  - **Contextual Hover Action Card:** Smoothly reveals a vibrant gradient button: *"Please provide an AI summary of this page."* complete with a `Sparkles` icon.
  - **1-Click Automated Research:** Clicking the hover button automatically dispatches an intelligent web page analysis prompt to the assistant.
  - **In-Thread Real-Time Stream:** The generated web summary streams seamlessly into the current thread as a new assistant turn, preserving workflow momentum.
  - **Debounced Non-Intrusive UX:** Engineered with a 250ms mouse exit delay and pointer safety margin to prevent UI flickering or accidental dismissal.

5 Archive Explorer & SQLite FTS5 Search

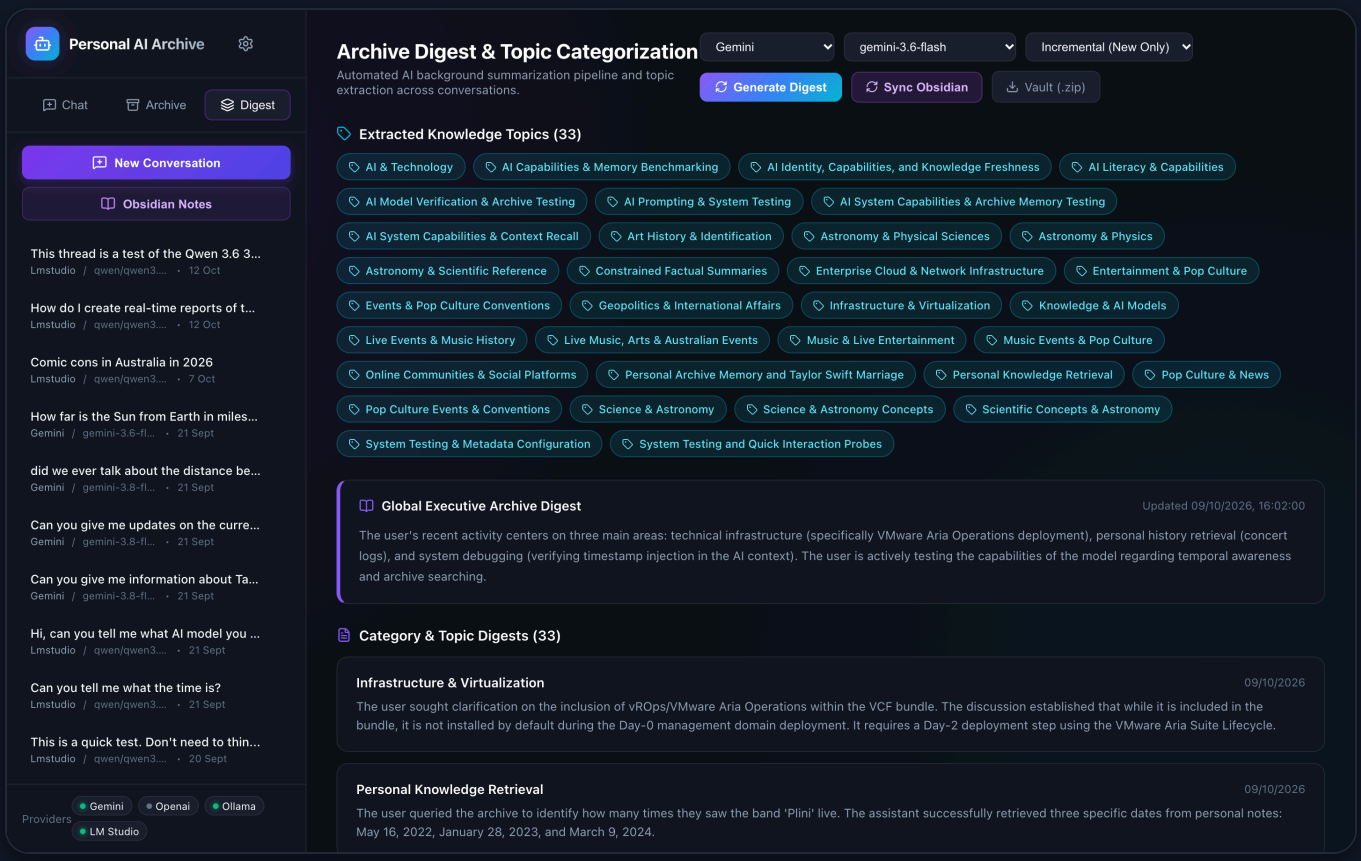
The centralized repository inspector powered by SQLite's FTS5 engine, providing instant token-based search across historical conversations, provider filtering chips, and export actions.



- **KEY ARCHITECTURAL FEATURES:**
  - **SQLite FTS5 Full-Text Search:** Instantaneous search bar indexing all user posts and assistant responses via database triggers for sub-millisecond retrieval.
  - **Multi-Provider Filter Chips:** One-click filtering by model vendor (`All` , `Gemini` , `OpenAI` , `Ollama` , `LM Studio`).
  - **Master-Detail Timeline View:** Left panel indexes conversation threads with message counters, dates, and provider badges; right preview pane displays the full transcript.
  - **Quick Thread Resumption:** Clicking any archived thread allows instant inspection or resuming active conversation with complete context continuity.

6 Intelligence Dashboard & Topic Categorization

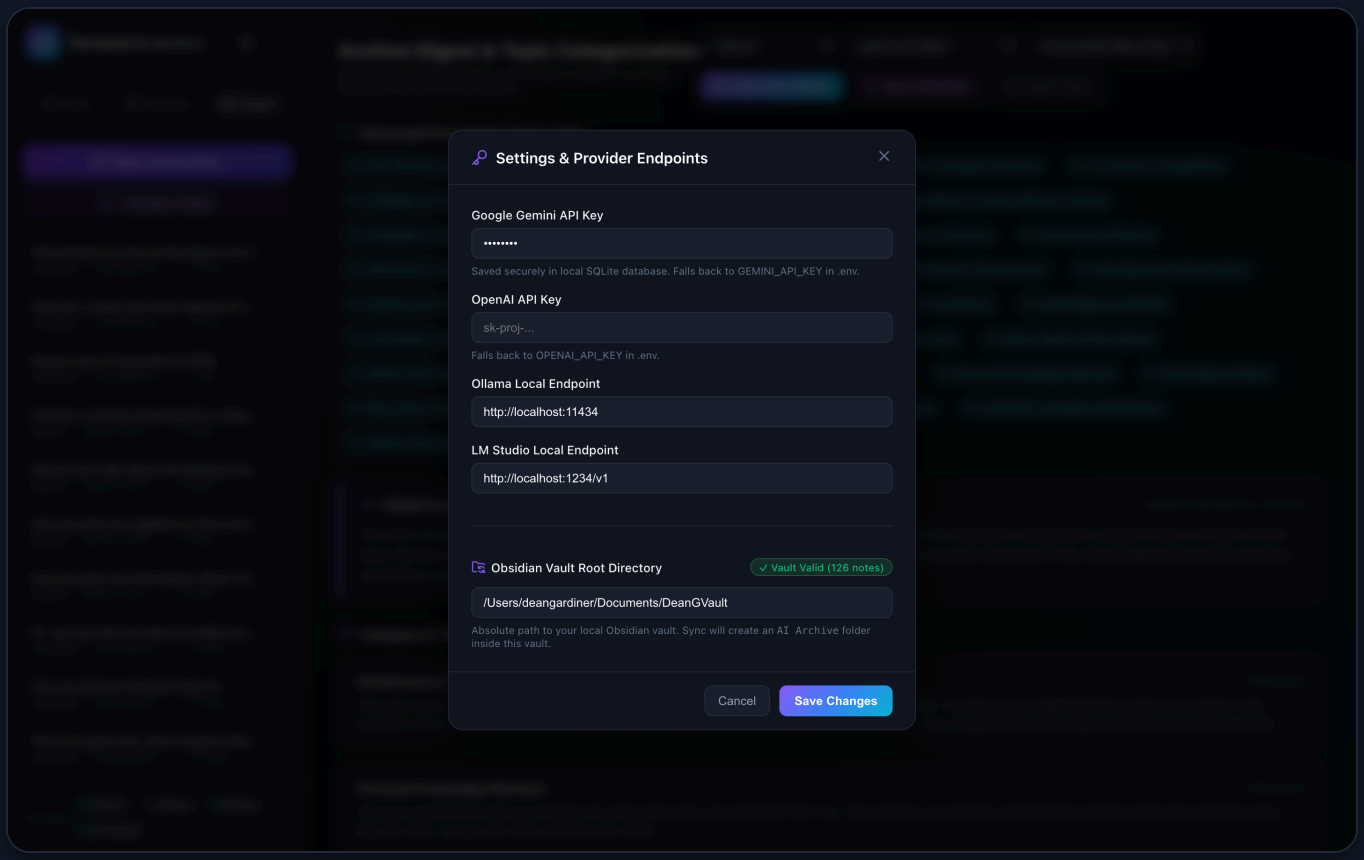
The automated background synthesis pipeline summarizing conversational history into global digests, automatically clustering recurring topics, and updating persistent Markdown records on disk.



- **KEY ARCHITECTURAL FEATURES:**
  - **Custom Provider & Model Selection:** Run synthesis on demand using cloud providers (Gemini, OpenAI) or 100% offline private local models (Ollama, LM Studio).
  - **Execution Modes:** Supports `Incremental (New Only)` mode for rapid updates and `Full Re-run (All History)` for comprehensive knowledge re-indexing.
  - **Extracted Knowledge Topics Grid:** Visual cards detailing automatically categorized knowledge domains, message tallies, and summarized themes.
  - **Global Executive Archive Digest:** Formatted briefing summarizing core project themes, decisions, active initiatives, and architectural evolutions.

7 Settings & Provider Endpoints Modal

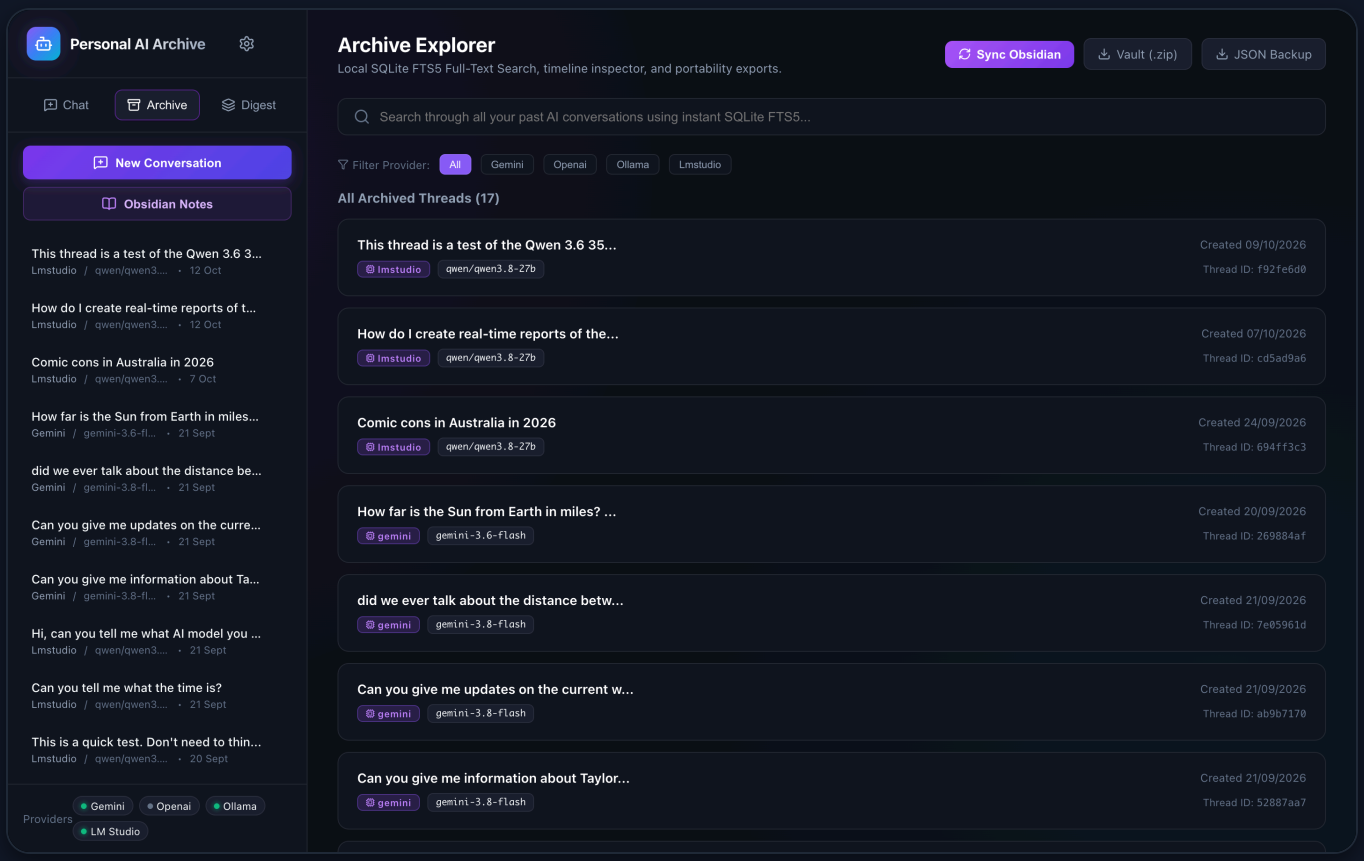
Configuration center for cloud API credentials, local offline AI server endpoints, and live Obsidian vault path verification, securely persisted in SQLite.



- **KEY ARCHITECTURAL FEATURES:**
  - **API Key Masking (`••••<last4>`):** Credentials for Gemini and OpenAI are safely masked upon retrieval, protecting secrets against screen shares and shoulder-surfing.
  - **Local Offline AI Endpoints:** Dedicated inputs for Ollama (`http://localhost:11434`) and LM Studio (`http://localhost:1234/v1`) with automatic model probe synchronization.
  - **Real-Time Debounced Vault Validation:** Validates user-specified Obsidian vault directory paths as typed, showing live index confirmations (`✓ Vault path valid (11 notes indexed)`).
  - **Relational Settings Persistence:** Settings persist in SQLite's `settings` table with intelligent fallback to environment variables in `.env`.

8 Data Portability Hub & Obsidian Vault Export

Guarantees complete user data ownership and zero vendor lock-in through streaming exports of structured Obsidian Markdown vaults, full JSON relational backups, and seamless cross-instance restoration with safe merge or clean clone options.



- **KEY ARCHITECTURAL FEATURES:**

- **Streaming Obsidian Vault (.zip) Export:** The `/api/export/markdown` endpoint packages all threads into standard Markdown files with YAML frontmatter, wikilinks, and organized folders.
- **Full Relational JSON Backup:** The `/api/export/json` endpoint provides a portable JSON snapshot containing complete conversation trees, messages, categories, and timestamps.
- **Cross-Instance JSON Restoration:** Interactive `RestoreModal` supporting both *Merge* (adds new conversations and updates matches without modifying existing threads) and *Clean Replace* modes with atomic SQLite transactions, missing-parent auto-healing, and sub-millisecond FTS5 re-indexing via `/api/restore/json`.
- **Continuous Two-Way Sync:** Direct synchronization into the user's active Obsidian vault directory (`DeanGVault/AI Archive`) ensures conversational learnings are instantly accessible in external knowledge graph tools.
- **Offline Data Sovereignty:** Complete local control over all SQLite databases, attachments, and exported archives with no cloud lock-in.